

# The Harmonic Mean

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## Introduction

The harmonic mean is the reciprocal of the arithmetic mean of reciprocals. It is appropriate when averaging rates, speeds, or ratios where the denominator is fixed – for example, when a traveller covers equal distances at different speeds and you want the average speed. In machine learning, the harmonic mean of precision and recall gives the F1 score.

## Prerequisites

The reader should know the arithmetic mean and be comfortable taking reciprocals in R.

## Theory

For positive values  $x_1, \dots, x_n$ ,

$$\text{HM}(x) = \frac{n}{\sum 1/x_i}.$$

The harmonic mean is always less than or equal to the geometric mean, which is less than or equal to the arithmetic mean:  $\text{HM} \leq \text{GM} \leq \text{AM}$ , with equality when all  $x_i$  are equal.

The F1 score in binary classification is the harmonic mean of precision and recall:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} = \frac{1}{\frac{1}{2} \left( \frac{1}{P} + \frac{1}{R} \right)} \cdot (\text{rescaled}).$$

## Assumptions

All values must be strictly positive. Negative values or zeros make the harmonic mean undefined or infinite.

## R Implementation

```
library(psych)

speeds <- c(40, 60, 80) # km/h over three equal distances
psych::harmonic.mean(speeds)
mean(speeds)             # incorrect for average speed over equal distances

# F1 score
precision <- 0.80
recall    <- 0.60
f1 <- 2 * precision * recall / (precision + recall)
f1

# Harmonic mean is appropriate for average rates
```

```
n <- 3
hm_speed <- n / sum(1 / speeds)
hm_speed
```

## Output & Results

```
[1] 55.38      # harmonic mean of speeds
[1] 60         # arithmetic mean (incorrect for equal-distance travel)
[1] 0.686      # F1 score with P=0.80, R=0.60
```

Travelling equal distances at 40, 60, 80 km/h gives an average speed of 55.4 km/h, not 60 km/h. The F1 of 0.69 splits the difference between P and R, penalising the lower value more than the arithmetic mean would.

## Interpretation

Use the harmonic mean when averaging rates with a fixed numerator or denominator (e.g., average speed when distance is constant). Use the F1 form when balancing two normalised scores where both values matter.

## Practical Tips

- Zero or negative inputs break the harmonic mean; guard against them.
- The harmonic mean weights small values more heavily; a single small observation drives the result down.
- In meta-analysis of rates per unit time, harmonic-mean sample size is sometimes used in effective-sample-size calculations.
- The F-beta score generalises F1:  $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$ , with  $\beta = 2$  emphasising recall and  $\beta = 0.5$  emphasising precision.
- Reporting a harmonic mean without explanation is rare; specify why it is the appropriate summary in the methods.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Descriptive statistics and Table 1